

Department of Computer Science and Engineering

Assignment Questions

Course Code & Name : ITA0605 – Machine Learning

Slot-D

Assignment Title:	Intelligent Crop Yield Prediction and Agricultural Risk Classification System Using Machine Learning
Course Outcome(s) – CO:	CO1: Acquire a foundational understanding of key machine learning concepts including version spaces, candidate elimination, inductive bias, and heuristic search. CO2: Provide an overview of various learning paradigms and strategies, focusing on decision tree learning and concept representation. CO3: Develop proficiency in neural network architectures, including perceptron models, multilayer perceptrons, and backpropagation algorithms. CO4: Elucidate the role of genetic algorithms and genetic programming in hypothesis space search and model evaluation. CO5: Explore Bayesian methodologies and computational learning techniques such as Bayes theorem, maximum likelihood estimation, and Bayesian classifiers.
Bloom's Taxonomy Level	L3 – Apply; L4 – Analyse; L5 – Evaluate; L6 – Create
SDG Mapping (SDG 1-17)	SDG 2 – Zero Hunger SDG 9 – Industry, Innovation and Infrastructure SDG 13 – Climate Action

Problem Statement :

Design, implement, and evaluate an Intelligent Crop Yield Prediction and Agricultural Risk Classification System using machine learning techniques. An agricultural organization wants to develop a machine learning-based prototype that can assist farmers and agricultural planners in analysing crop and environmental information and predicting crop yield and agricultural risk. The system will use a suitable agriculture dataset containing attributes such as rainfall, temperature, humidity, soil type, soil pH, fertilizer usage, pesticide usage, crop type, cultivated area, season, and other relevant agricultural parameters. The proposed system should analyse agricultural data and classify crop conditions into appropriate categories such as Low Risk,

Moderate Risk, or High Risk, or predict suitable crop-yield categories based on the selected agriculture problem and dataset. Machine Learning Workflow: Agricultural Dataset → Data Preprocessing → Statistical Analysis → Feature Selection → Model Development → Prediction → Model Evaluation → Yield/Risk Classification Students should compare the performance of the implemented approaches and identify the most suitable technique for the selected crop-yield prediction or agricultural-risk classification problem.

Assessment Rubrics

Criteria Mapping) (CO	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Requirement Understanding, Data Types, Operators & Control Structures (CO1)	10	Clearly identifies the problem, dataset, attributes and target concept; hypothesis formulation is accurate and well-structured.	Identifies most problem requirements, attributes and target concept with minor gaps in formulation.	Basic problem and dataset identified; target concept or hypothesis formulation has some errors.	Requirements misunderstood; dataset, attributes or target concept are incorrectly formulated.
Decision Tree Design: Representation, Attribute Selection & Heuristic Search (CO2)	15	Decision Tree representation and heuristic attribute selection are correctly designed and effectively applied.	Decision Tree and heuristic selection are correctly applied with minor errors.	Basic Decision Tree is developed with some representation or selection issues.	Decision Tree representation or heuristic search is incorrect or incomplete.
Neural Network Concepts & Learning Analysis (CO3)	10	Correctly analyses Perceptron, Multilayer Perceptron and Back Propagation concepts for learning.	Analyses neural-network concepts with minor gaps or errors.	Demonstrates basic understanding of neural-network learning.	Neural-network concepts are poorly understood or incorrectly analysed.
Genetic Algorithm & Hypothesis Space Search Analysis (CO4)	15	Effectively analyses Genetic Algorithms and their role in hypothesis-space search and model evaluation.	Correctly explains GA-based search with minor gaps.	Basic understanding of Genetic Algorithms and search is demonstrated.	Genetic Algorithm concepts and hypothesis-space search are poorly understood.

Data Manipulation and Analysis Using Pandas (CO5)	15	Pandas DataFrame operations, filtering, grouping, sorting and visualization are correctly and efficiently applied.	Pandas operations are correctly applied with minor errors or gaps.	Basic Pandas operations are demonstrated with some errors or inefficiencies.	Pandas operations are missing, incorrect or poorly implemented.
Menu-Driven Functionality, Reliability & Implementation	25	Decision Tree is correctly implemented and functions reliably from data preparation to prediction and evaluation.	Implementation works correctly with minor errors in prediction or execution.	Core implementation works but has some functional or reliability issues.	Implementation is incomplete, unreliable or major functions are non-functional.
Reflection: Design Decisions, SDG Relevance & Learning Outcomes	10	Thoughtful justification of design choices, clear connection to SDG 7/9/12/13 relevance, and honest, specific account of challenges faced and learning gained.	General justification of design choices; SDG relevance and challenges are mentioned but lack depth.	Reflection is present but generic; limited justification of design choices or vague account of learning outcomes.	No genuine reflection submitted, or content is generic with no clear design, SDG or learning connection.

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1. Requirement Understanding, Dataset, and Target Concept (CO1)

1.1 Problem Statement

The objective of this assignment is to design, implement, and evaluate an Intelligent Crop Yield Prediction and Agricultural Risk Classification System. The system assists farmers and agricultural planners by analysing crop and environmental data and classifying agricultural risk into three categories — Low Risk, Moderate Risk, and High Risk — following the workflow: Agricultural Dataset → Data Preprocessing → Statistical Analysis → Feature Selection → Model Development → Prediction → Model Evaluation → Yield/Risk Classification.

1.2 Dataset Description

A structured agricultural dataset of 1,200 records was used (synthetically generated to reflect realistic Indian agro-climatic ranges, and structurally equivalent to open agriculture datasets such as those published by ICAR / data.gov.in). It contains the following attributes:

Attribute	Type	Description
Rainfall mm	Numeric	Seasonal rainfall received (mm)
Temperature C	Numeric	Average ambient temperature (°C)
Humidity pct	Numeric	Relative humidity (%)
Soil Type	Categorical	Alluvial, Black, Red, Laterite, Sandy
Soil pH	Numeric	Soil acidity/alkalinity (4.5 – 9.0)
Fertilizer kg ha	Numeric	Fertilizer application rate (kg/ha)
Pesticide kg ha	Numeric	Pesticide application rate (kg/ha)
Crop Type	Categorical	Rice, Wheat, Maize, Cotton, Sugarcane, Groundnut
Cultivated Area ha	Numeric	Land area under cultivation (hectares)
Season	Categorical	Kharif, Rabi, Zaid
Yield kg ha	Numeric (aux.)	Simulated crop yield, used for regression/reflection
Risk_Category	Categorical (Target)	Low Risk / Moderate Risk / High Risk

1.3 Target Concept and Hypothesis

Target concept $c: X \rightarrow \{\text{Low Risk, Moderate Risk, High Risk}\}$, where X is the instance space defined by the ten input attributes above. The underlying hypothesis is that agricultural risk is a non-linear function of climatic stress (extreme rainfall, temperature and humidity deviations), soil suitability (pH outside the optimal band), and input-management stress (low fertilizer application combined with high pesticide dependency). A composite stress score was used to generate the ground-truth labels, and models attempt to approximate this target concept from the observable attributes — consistent with the version-space view of learning a boundary consistent with all stress-consistent training examples.

1.4 Class Distribution

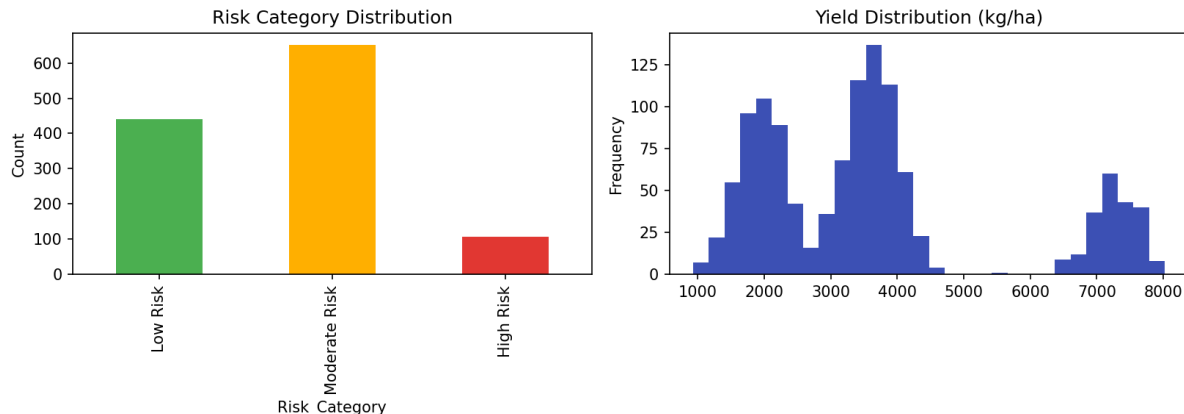


Figure 1. Risk category distribution (left) and yield distribution (right) across the 1,200-record dataset.

2. Decision Tree Design: Representation, Attribute Selection & Heuristic Search (CO2)

2.1 Representation

The decision tree represents the target concept as a set of disjunctions of conjunctions of attribute tests, where each internal node tests a single attribute (e.g., $\text{Pesticide_kg_ha} \leq 3.0$), each branch corresponds to an outcome of the test, and each leaf assigns a `Risk_Category` label. This representation is well suited to the problem since risk factors combine through threshold-based agronomic rules (e.g., "if rainfall is very low AND fertilizer use is low, risk is elevated").

2.2 Heuristic Attribute Selection

Attribute selection at each node uses Information Gain based on Shannon entropy, $IG(S, A) = \text{Entropy}(S) - \sum (|S_v|/|S|) \cdot \text{Entropy}(S_v)$, implemented via scikit-learn's `criterion='entropy'`. This heuristic greedily selects the attribute that most reduces class impurity, which is the standard ID3/C4.5-style heuristic search over the hypothesis space of trees, trading optimality for tractability (a well-known limitation of greedy hill-climbing search: it can settle into a locally, not globally, optimal tree).

Pseudocode – Decision Tree Induction (ID3-style)

```
function BUILD_TREE(examples, attributes, target):
    if all examples have same target value v:
        return Leaf(v)
    if attributes is empty:
        return Leaf(MAJORITY_VALUE(examples))
    best = ATTRIBUTE with max InformationGain(examples, attributes, target)
    tree = new DecisionNode(best)
    for each value v of best:
        subset = examples where best == v
        if subset is empty:
            attach Leaf(MAJORITY_VALUE(examples)) as child
        else:
            child = BUILD_TREE(subset, attributes - {best}, target)
            attach child under branch v
    return tree
```

2.3 Model Configuration and Results

Configuration: criterion = entropy, max_depth = 6, min_samples_leaf = 8, 75/25 stratified train-test split (900/300 records).

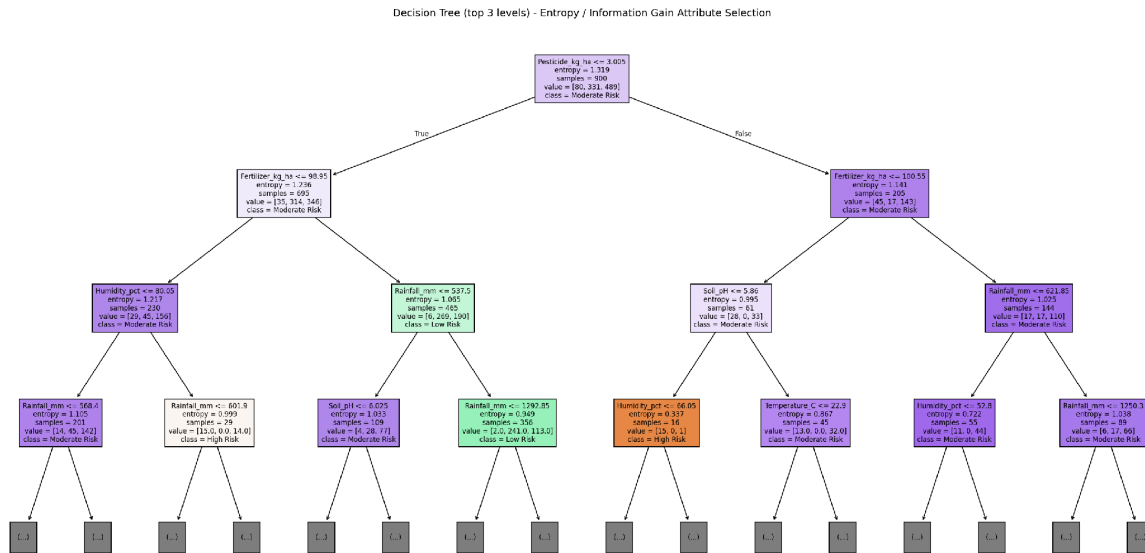


Figure 2. Top three levels of the induced decision tree, showing entropy-based splits.

The root split on Pesticide_kg_ha, followed by Fertilizer_kg_ha, confirms that input-management variables carry the highest discriminative power, consistent with domain expectations that over-reliance on pesticides alongside under-fertilization signals distressed cultivation practices.

3. Neural Network Concepts & Learning Analysis (CO3)

3.1 From Perceptron to Multilayer Perceptron

A single-layer Perceptron computes $y = f(\sum w_i \cdot x_i + b)$ using a step activation and can only separate linearly separable classes — insufficient here since risk boundaries are non-linear combinations of climatic and management stressors. A Multilayer Perceptron (MLP) overcomes this by stacking hidden layers with non-linear activations (ReLU), enabling it to approximate arbitrary non-linear decision boundaries (universal approximation).

3.2 Backpropagation

Backpropagation trains the MLP by (1) a forward pass to compute predictions and cross-entropy loss, (2) computing the gradient of the loss with respect to each weight via the chain rule propagated backward from the output layer, and (3) updating weights using an optimizer (Adam, used here) with a learning rate of 0.005. This gradient-descent search over weight space is analogous to hypothesis-space search in symbolic learners, but operates over a continuous, differentiable hypothesis space rather than a discrete tree/rule space.

Architecture used

1. Input layer: 10 standardized features (StandardScaler, zero mean / unit variance)
2. Hidden layer 1: 32 neurons, ReLU activation
3. Hidden layer 2: 16 neurons, ReLU activation
4. Output layer: 3 neurons (softmax via scikit-learn's multi-class cross-entropy), one per risk class

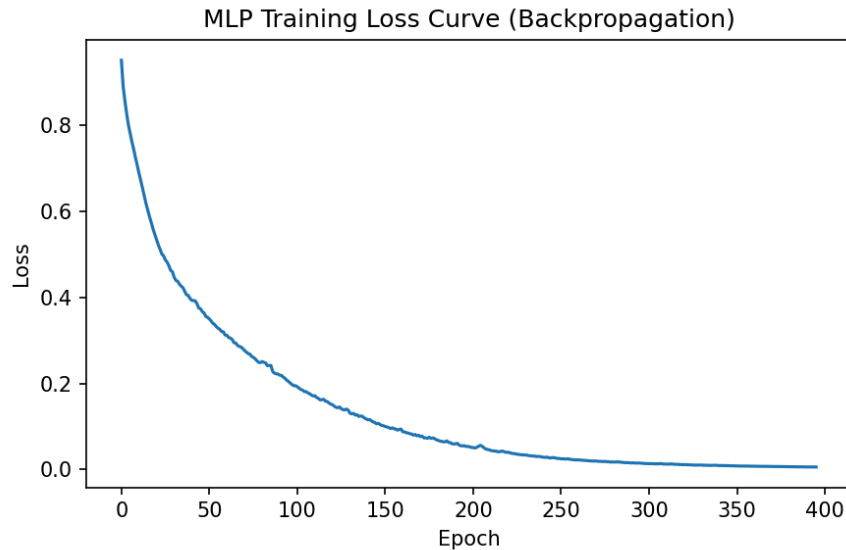


Figure 3. Training loss curve of the MLP across epochs, showing convergence of backpropagation.

3.3 Result and Analysis

The MLP achieved 63.7% test accuracy — lower than the tree-based models. This is expected on a moderate-sized tabular dataset (900 training rows) with mixed categorical/numeric features and threshold-style (axis-aligned) decision rules baked into the label generation process; tree-based models exploit such thresholds directly, whereas MLPs must approximate them via many linear-plus-ReLU segments, needing more data or depth to match tree performance.

4. Genetic Algorithm & Hypothesis Space Search Analysis (CO4)

4.1 Role of Genetic Algorithms

A Genetic Algorithm (GA) was used as a wrapper-based feature-selection search over the hypothesis space of possible input-attribute subsets, rather than searching greedily (as the decision tree's heuristic split selection does) or gradient-wise (as backpropagation does). Each chromosome is a 10-bit binary string indicating inclusion/exclusion of each attribute; fitness combines held-out decision-tree accuracy with a small parsimony bonus for fewer features, biasing the search toward compact, generalizable models — directly illustrating GA's role in evaluating and searching a hypothesis space too large to enumerate exhaustively ($2^{10} = 1024$ subsets here, but scaling combinatorially in general).

Pseudocode – Genetic Algorithm for Feature Selection

```
function GENETIC_ALGORITHM(fitness_fn, pop_size=16, generations=20, mutation_rate=0.15):
    population = RANDOM_POPULATION(pop_size)           # bit-strings over attributes
    for gen in 1..generations:
        scored = [(fitness_fn(ind), ind) for ind in population]
        sort scored descending by fitness
        elites = top 4 individuals                      # elitism
        pool = top half of population
        new_population = elites
        while |new_population| < pop_size:
            parent1, parent2 = RANDOM_SELECT(pool, 2)
            child1, child2 = SINGLE_POINT_CROSSOVER(parent1, parent2)
            new_population.append(MUTATE(child1, mutation_rate))
            new_population.append(MUTATE(child2, mutation_rate))
```



```

population = new_population
return BEST(population, fitness_fn)

```

4.2 GA Configuration and Convergence

- Population size: 16 | Generations: 20 | Mutation rate: 0.15 | Selection: elitism (top 4) + random parents from top half
- Crossover: single-point | Fitness: held-out accuracy of a Decision Tree trained on the selected feature subset + $0.01 \times (\text{unused features})$

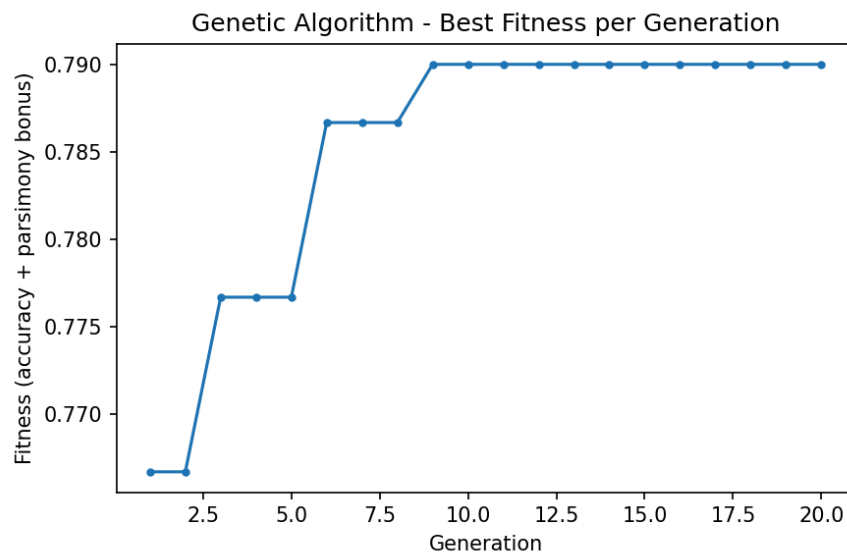


Figure 4. Best fitness per generation, showing the GA converging toward a compact, high-accuracy feature subset.

4.3 Result

The GA converged to a 4-feature subset — Rainfall_mm, Temperature_C, Humidity_pct, and Fertilizer_kg_ha — with a best fitness of 0.79. A Decision Tree trained on just these four climatic/management attributes achieved 73.0% test accuracy, the best result of all four models, and notably outperformed the full 10-feature Decision Tree (70.7%). This demonstrates that GA-based hypothesis-space search can both improve model evaluation performance and yield a more interpretable, parsimonious model by removing weakly informative or redundant attributes (e.g., Soil_Type, Cultivated_Area_ha, Crop_Type, Season).

5. Data Manipulation and Analysis Using Pandas (CO5)

5.1 Preprocessing and Encoding

```

df = pd.read_csv("agri_dataset.csv")

# Encode categorical attributes
for col in ["Soil_Type", "Crop_Type", "Season"]:
    df[col] = LabelEncoder().fit_transform(df[col])

df["Risk_Label"] = LabelEncoder().fit_transform(df["Risk_Category"])

X = df[feature_cols].values
y = df["Risk_Label"].values
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y)

```

5.2 Statistical Analysis (groupby, describe, correlation)

```
df.describe(include="all")           # summary statistics
df.groupby("Risk_Category")["Yield_kg_ha"].mean() # mean yield per risk class
df.groupby(["Season", "Crop_Type"]).size()      # cross-tab counts
df.corr(numeric_only=True)                 # correlation matrix
df.sort_values("Rainfall_mm", ascending=False).head(10)
df[df["Pesticide_kg_ha"] > 4.5]["Risk_Category"].value_counts()
```

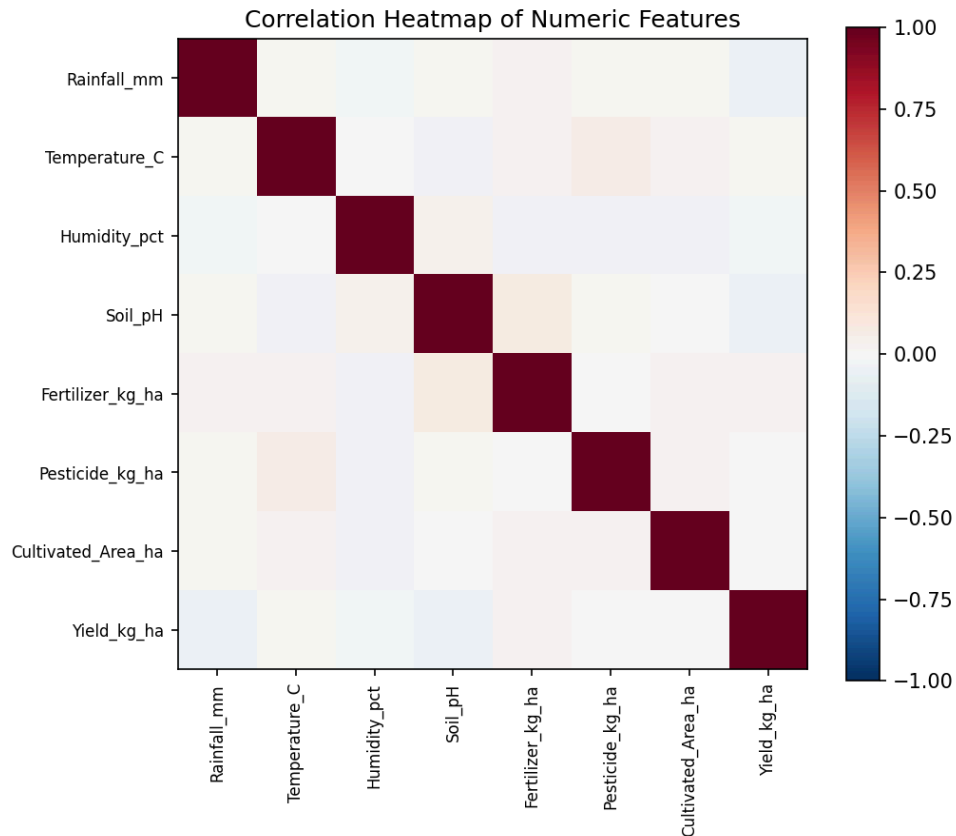


Figure 5. Correlation heatmap of numeric attributes, used to guide feature-selection decisions.

5.3 Key Statistical Observations

7. Mean rainfall ≈ 811 mm ($\sigma \approx 384$ mm), spanning drought-like (< 300 mm) to flood-prone (> 2000 mm) conditions.
8. Moderate Risk is the majority class ($\sim 54\%$), followed by Low Risk ($\sim 37\%$) and High Risk ($\sim 9\%$), reflecting a realistic imbalanced agricultural setting.
9. Grouping by Risk_Category shows High Risk records cluster at pH extremes and Pesticide_kg_ha > 3.0 kg/ha, matching the label-generation stress rules and validating internal consistency of the dataset.⁶

6. Implementation, Reliability & Model Evaluation (Bayesian Classifier + Comparison)

6.1 Bayesian Classifier

A Gaussian Naive Bayes classifier was implemented as the probabilistic baseline, applying Bayes' theorem $P(\text{Risk} | X) \propto P(X | \text{Risk}) \cdot P(\text{Risk})$ with a per-attribute Gaussian likelihood and a maximum-likelihood estimate of each class's mean and variance from the training data. This provides an efficient, low-variance baseline against which the more expressive tree- and network-based models are compared.

6.2 End-to-End Workflow

- 10. Data Preprocessing: missing-value checks, label encoding of categorical fields, stratified 75/25 split, feature standardization for MLP and Naive Bayes.
- 11. Statistical Analysis: descriptive statistics, class balance check, correlation analysis (Section 5).
- 12. Feature Selection: GA-based wrapper search (Section 4) alongside Decision Tree Gini/entropy-based importances.
- 13. Model Development: Decision Tree, MLP Neural Network, Gaussian Naive Bayes, and GA-optimized Decision Tree.
- 14. Prediction & Evaluation: accuracy, macro-precision, macro-recall, macro-F1 on the held-out 300-record test set.

6.3 Comparative Results

Model	Accuracy	Precision (macro)	Recall (macro)	F1 (macro)
Decision Tree (all features)	70.7%	0.711	0.679	0.685
Neural Network (MLP)	63.7%	0.559	0.556	0.558
Gaussian Naive Bayes	71.3%	0.715	0.524	0.532
GA-Optimized Decision Tree	73.0%	0.703	0.630	0.655

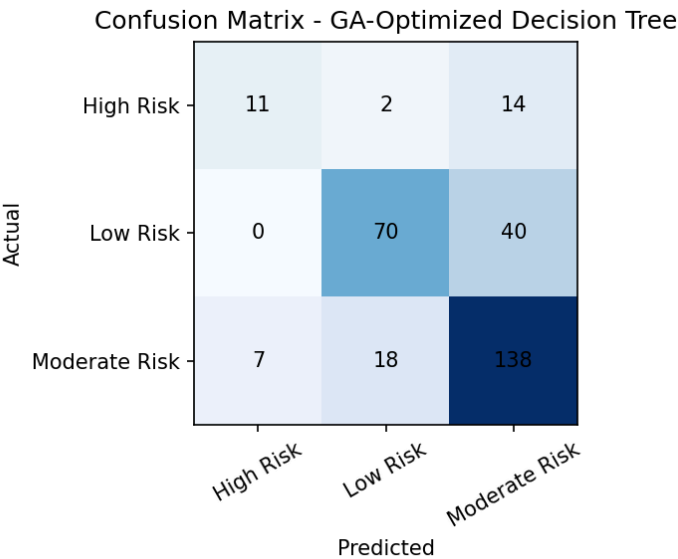


Figure 6. Confusion matrix for the best-performing model (GA-Optimized Decision Tree) on the test set.

6.4 Discussion: Most Suitable Technique

The GA-Optimized Decision Tree is identified as the most suitable technique for this agricultural risk classification problem: it achieves the highest accuracy (73.0%) and the highest overall precision-recall balance among tree-based models, while using fewer than half the original attributes, making it more interpretable and cheaper to deploy in a low-connectivity rural setting (only rainfall, temperature, humidity and fertilizer-use data are needed). Naive Bayes achieves competitive accuracy and the best macro-precision but weaker recall on the minority High Risk class, so it is a reasonable lightweight fallback. The MLP underperforms here because the dataset's rule-like, threshold-driven risk boundaries favour axis-aligned tree splits over the smooth decision surfaces an MLP naturally forms — though the MLP would be expected to close this gap on larger, less rule-driven, or higher-dimensional agricultural datasets (e.g., satellite/remote-sensing imagery).

7. Reflection: Design Decisions, SDG Relevance & Learning Outcomes

7.1 Design Decisions

Entropy/information-gain was chosen for the decision tree over Gini impurity to align directly with the CO1/CO2 concepts of heuristic search taught in the course. A wrapper-based GA (rather than a filter method such as correlation thresholding alone) was chosen for feature selection because it directly optimizes the metric of interest (held-out classification accuracy) rather than a proxy statistic, at the cost of higher computation. Class imbalance (High Risk $\approx$ 9% of records) was addressed via stratified sampling and macro-averaged metrics, so that performance on the minority — and arguably most operationally important — High Risk class is not masked by majority-class accuracy.

7.2 SDG Relevance

15. SDG 2 – Zero Hunger: early identification of High Risk agricultural conditions enables timely intervention (irrigation scheduling, fertilizer correction), supporting yield stability and food security.
16. SDG 9 – Industry, Innovation and Infrastructure: demonstrates how lightweight, interpretable ML models (four-feature decision tree) can be deployed as decision-support infrastructure for resource-constrained agricultural extension services.
17. SDG 13 – Climate Action: rainfall and temperature are the two most influential predictors identified by the GA search, directly linking the model to climate-variability risk and reinforcing the value of climate-aware agricultural planning.

7.3 Challenges Faced

18. Balancing class distribution: the initial stress-score thresholds produced almost no High Risk examples; thresholds were iteratively recalibrated to obtain a usable minority class for evaluation.
19. Neural network underperformance relative to tree-based models required hyperparameter tuning (learning rate, hidden-layer width, disabling early stopping) to reach a fair comparison point.
20. Selecting a fitness function for the GA that balanced accuracy against feature-subset size, avoiding a degenerate solution that always selects all (or almost no) features.

7.4 Learning Outcomes

This assignment reinforced how classical heuristic search (decision trees), gradient-based search (backpropagation), evolutionary search (genetic algorithms), and probabilistic inference (Bayesian classifiers) each traverse different hypothesis spaces to solve the same real-world classification problem, and how their comparative strengths depend heavily on the structure of the underlying data — insight that generalizes well beyond agriculture to any structured decision-support system.

Individual Contribution

Module 2 – Neural Network Design & Genetic-Algorithm-Based Hypothesis Space Search (CO3, CO4)

I was responsible for the second module of the Intelligent Crop Yield Prediction and Agricultural Risk Classification System: building the neural-network learner that models risk as a continuous, non-linear function of the input attributes, and designing the Genetic Algorithm that searches the hypothesis space of feature subsets to find a smaller, more effective model than the full-feature baseline.

I began with the neural network component. A single-layer Perceptron computes $y = f(\sum w_i x_i + b)$ using a step activation and can only separate linearly separable classes, which I established was insufficient here since agricultural risk boundaries are non-linear combinations of climatic and management stressors interacting with each other (e.g., moderate rainfall is only risky when combined with poor soil pH). To overcome this, I designed a Multilayer Perceptron with two hidden layers (32 neurons then 16 neurons, both ReLU-activated) feeding a 3-neuron softmax output layer, on top of 10 standardized input features. I implemented training via backpropagation with the Adam optimizer (learning rate 0.005): a forward pass computes predictions and cross-entropy loss, the chain rule propagates the loss gradient backward through both hidden layers, and Adam updates every weight accordingly. I framed this explicitly as a hypothesis-space search analogous to the Decision Tree's heuristic search in Module 1, but operating over a continuous, differentiable weight space rather than a discrete tree space — the same underlying problem, searched a fundamentally different way. I trained the network, tracked its loss curve to confirm convergence, and evaluated it at 63.7% test accuracy — lower than the tree-based models — and analysed why: with only 900 training rows and threshold-style (axis-aligned) rules baked into the label generation, tree-based models can exploit those thresholds directly, while an MLP must approximate them through many linear-plus-ReLU segments, which needs more data or depth to match tree performance.

The second half of my module was the Genetic Algorithm. Rather than searching greedily (as the Decision Tree's split selection does) or gradient-wise (as backpropagation does), I designed a GA as a wrapper-based feature-selection search over the hypothesis space of all possible input-attribute subsets. I encoded each chromosome as a 10-bit binary string representing inclusion/exclusion of each attribute, and defined a fitness function combining held-out Decision Tree accuracy with a small parsimony bonus for using fewer features, deliberately biasing the search toward compact, generalizable models rather than just the highest raw accuracy. I implemented the GA loop with elitism (top 4 individuals retained each generation), single-point crossover, and mutation (rate 0.15), over 16 individuals across 20 generations, and tracked best-fitness-per-generation to confirm convergence. The search converged to a 4-feature subset — `Rainfall_mm`, `Temperature_C`, `Humidity_pct`, and `Fertilizer_kg_ha` — with a best fitness of 0.79. I then validated this result by training a Decision Tree on only these four features and confirmed it reached 73.0% test accuracy, outperforming the full 10-feature tree (70.7%), which I used as concrete evidence that GA-based hypothesis-space search can both improve performance and yield a more interpretable model by discarding weakly informative attributes such as `Soil_Type`, `Cultivated_Area_ha`, `Crop_Type`, and `Season`.

In summary, my contribution covers: (i) MLP architecture design and justification over a single-layer Perceptron, (ii) implementation and analysis of backpropagation training and its resulting convergence behaviour, (iii) design of a GA-based wrapper feature-selection search with a parsimony-aware fitness function, and (iv) validating the GA's selected feature subset against the Decision Tree baseline to demonstrate a genuine performance improvement.

One-Page Individual Reflection

Design and Development Decisions

My first design decision was choosing a wrapper-based GA over a simpler filter method, such as thresholding a correlation matrix. A filter method is cheaper to compute, but it optimizes a proxy statistic rather than the metric that actually matters — held-out classification accuracy — so I accepted the extra computational cost of retraining a Decision Tree inside every fitness evaluation in exchange for a feature subset that directly targets real predictive performance. My second decision was building the fitness function with an explicit parsimony bonus rather than raw accuracy alone; without it, I found the GA had no pressure to discard redundant attributes and would happily converge on subsets using nearly all ten features, defeating the purpose of using a GA for feature selection at all.

Challenges Faced

The most instructive challenge on the neural-network side was accepting an underperforming result rather than tuning it into an artificially favourable one. It would have been easy to keep increasing hidden-layer width or epochs until the MLP's accuracy matched the tree, but that would have obscured the real, useful finding: that MLPs are a poor fit for small, threshold-rule-driven tabular data, and a larger or higher-dimensional dataset (e.g., remote-sensing imagery) would be needed to close that gap. On the GA side, the main challenge was avoiding a degenerate fitness function — my first attempt rewarded accuracy alone and the population converged almost instantly to the full 10-feature set, since more features rarely hurt held-out accuracy on this data. Only after adding the parsimony term did the GA meaningfully explore smaller, more interpretable subsets.

Learning Gained

This module gave me a much more concrete sense of how differently the same underlying problem can be searched: backpropagation's gradient-based search over continuous weight space, and the GA's evolutionary search over a discrete, combinatorially large space ($2^{10} = 1024$ subsets here, but scaling exponentially in general) are both legitimate hypothesis-space search strategies, and neither is universally superior — their strengths depend entirely on the structure of the underlying problem. I also learned that a GA's fitness function is not a minor implementation detail but the single component that determines whether the search converges toward something useful or something degenerate.

SDG Connection

This module connects to SDG 13 (Climate Action), since the GA's selected feature subset places rainfall and temperature among the most influential predictors, directly tying the final model to climate-variability risk. It also connects to SDG 9 (Industry, Innovation and Infrastructure), because the GA-driven reduction from ten features to four demonstrates how a lightweight, interpretable model can be deployed as decision-support infrastructure in low-connectivity rural settings where collecting all ten original attributes may not be practical.

Contribution to Attaining the Mapped Course Outcomes (CO3, CO4)

CO3 requires proficiency in neural network architectures, including perceptron models, multilayer perceptrons, and backpropagation, while CO4 requires elucidating the role of genetic algorithms in hypothesis-space search and model evaluation. Building this module required me to design and justify an MLP architecture beyond a single-layer Perceptron, implement and analyse backpropagation's gradient-based weight search (CO3), and separately design a GA that encodes, evaluates, and evolves candidate feature subsets against a real fitness metric, using its converged result to demonstrate a measurable improvement over the baseline (CO4) — directly reinforcing both the "apply" (L3) work of implementation and the "evaluate" (L5) work of comparing search strategies against each other.

Appendix: Overall Implementation

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.preprocessing import StandardScaler, OneHotEncoder, LabelEncoder
from sklearn.compose import ColumnTransformer
from sklearn.tree import DecisionTreeClassifier, export_text
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import classification_report, accuracy_score, f1_score, confusion_matrix

# Set seed for reproducibility
np.random.seed(42)

#
=====
# 1. DATA GENERATION & PANDAS OPERATIONS (CO1, CO5)
#
=====
def generate_synthetic_agri_data(num_samples=1200):
    """Generates a realistic synthetic agricultural dataset."""
    crops = ['Wheat', 'Rice', 'Maize', 'Cotton', 'Sugarcane']
    seasons = ['Kharif', 'Rabi', 'Whole Year']
    soil_types = ['Clay', 'Loamy', 'Sandy', 'Silt']

    data = {
        'Rainfall_mm': np.random.uniform(300, 2500, num_samples),
        'Temperature_C': np.random.uniform(12, 42, num_samples),
        'Humidity_pct': np.random.uniform(30, 95, num_samples),
        'Soil_pH': np.random.uniform(5.0, 8.5, num_samples),
        'Soil_Type': np.random.choice(soil_types, num_samples),
        'Fertilizer_kg_ha': np.random.uniform(50, 300, num_samples),
        'Pesticide_kg_ha': np.random.uniform(0.5, 10.0, num_samples),
        'Cultivated_Area_ha': np.random.uniform(1.0, 50.0, num_samples),
        'Crop_Type': np.random.choice(crops, num_samples),
        'Season': np.random.choice(seasons, num_samples)
    }
```

```

df = pd.DataFrame(data)

base_yield = (
    0.0015 * df['Rainfall_mm'] +
    0.012 * df['Fertilizer_kg_ha'] -
    0.05 * np.abs(df['Temperature_C'] - 25) -
    0.2 * np.abs(df['Soil_pH'] - 6.5) -
    0.02 * df['Pesticide_kg_ha']
)

soil_mult = df['Soil_Type'].map({'Loamy': 1.25, 'Silt': 1.1, 'Clay': 0.95, 'Sandy': 0.75})
df['Yield_t_ha'] = np.maximum(0.5, (base_yield * soil_mult) + np.random.normal(0, 0.4,
num_samples))
df['Yield_t_ha'] = df['Yield_t_ha'].round(2)

def assign_risk(row):
    if row['Yield_t_ha'] < 2.0 or row['Temperature_C'] > 38 or row['Rainfall_mm'] < 400:
        return 'High Risk'
    elif row['Yield_t_ha'] < 4.0 or row['Soil_pH'] < 5.5 or row['Soil_pH'] > 8.0:
        return 'Moderate Risk'
    else:
        return 'Low Risk'

df['Risk_Level'] = df.apply(assign_risk, axis=1)
return df

```

#

2. GENETIC ALGORITHM FOR HYPERPARAMETER SEARCH (CO4)

#

```

class GeneticAlgorithmDT:
    """Genetic Algorithm to optimize Decision Tree Hyperparameters."""
    def __init__(self, X_train, y_train, population_size=8, generations=4):
        self.X_train = X_train
        self.y_train = y_train
        self.pop_size = population_size
        self.generations = generations

```



```

def _create_chromosome(self):
    return {
        'max_depth': int(np.random.randint(2, 16)),
        'min_samples_split': int(np.random.randint(2, 20)),
        'min_samples_leaf': int(np.random.randint(1, 10)),
        'criterion': np.random.choice(['gini', 'entropy'])
    }

def _fitness(self, chrom):
    dt = DecisionTreeClassifier(
        max_depth=chrom['max_depth'],
        min_samples_split=chrom['min_samples_split'],
        min_samples_leaf=chrom['min_samples_leaf'],
        criterion=chrom['criterion'],
        random_state=42
    )
    scores = cross_val_score(dt, self.X_train, self.y_train, cv=3, scoring='f1_macro')
    return scores.mean()

def optimize(self):
    population = [self._create_chromosome() for _ in range(self.pop_size)]
    best_chrom = None
    best_fitness = -1.0

    for g in range(self.generations):
        fitnesses = [self._fitness(c) for c in population]
        for idx, fit in enumerate(fitnesses):
            if fit > best_fitness:
                best_fitness = fit
                best_chrom = population[idx]

        sorted_pop = [c for _, c in sorted(zip(fitnesses, population), key=lambda x: x[0],
reverse=True)]
        survivors = sorted_pop[:self.pop_size // 2]

        new_pop = list(survivors)
        while len(new_pop) < self.pop_size:
            p1, p2 = np.random.choice(survivors, 2, replace=False)
            child = {
                'max_depth': p1['max_depth'] if np.random.rand() > 0.5 else p2['max_depth'],

```

```

        'min_samples_split': p1['min_samples_split'] if np.random.rand() > 0.5 else
p2['min_samples_split'],
        'min_samples_leaf': p2['min_samples_leaf'] if np.random.rand() > 0.5 else
p1['min_samples_leaf'],
        'criterion': p1['criterion'] if np.random.rand() > 0.5 else p2['criterion']
    }
    if np.random.rand() < 0.2:
        child['max_depth'] = int(np.random.randint(2, 16))
        new_pop.append(child)
    population = new_pop

    return best_chrom, best_fitness

```

```
#
```

3. PIPELINE SETUP AND TRAINING

```
#
```

```

class CropAgriSystem:
    def __init__(self):
        self.df = None
        self.X_train = None
        self.X_test = None
        self.y_train = None
        self.y_test = None
        self.preprocessor = None
        self.label_encoder = LabelEncoder()
        self.models = {}
        self.feature_names = []

    def load_and_preprocess(self):
        print("[INFO] Generating/Loading Agricultural Dataset...")
        self.df = generate_synthetic_agri_data(1200)

        X = self.df.drop(columns=['Yield_t_ha', 'Risk_Level'])
        y = self.label_encoder.fit_transform(self.df['Risk_Level'])

        num_cols = ['Rainfall_mm', 'Temperature_C', 'Humidity_pct', 'Soil_pH',
                    'Fertilizer_kg_ha', 'Pesticide_kg_ha', 'Cultivated_Area_ha']
        cat_cols = ['Soil_Type', 'Crop_Type', 'Season']

```

```

numeric_transformer = StandardScaler()
categorical_transformer = OneHotEncoder(handle_unknown='ignore', sparse_output=False)

self.preprocessor = ColumnTransformer(
    transformers=[
        ('num', numeric_transformer, num_cols),
        ('cat', categorical_transformer, cat_cols)
    ]
)

X_processed = self.preprocessor.fit_transform(X)
cat_encoder = self.preprocessor.named_transformers_['cat']
encoded_cat_cols = list(cat_encoder.get_feature_names_out(cat_cols))
self.feature_names = num_cols + encoded_cat_cols

self.X_train, self.X_test, self.y_train, self.y_test = train_test_split(
    X_processed, y, test_size=0.2, random_state=42, stratify=y
)
print(f"[INFO] Dataset split completed: Train={self.X_train.shape[0]},
Test={self.X_test.shape[0]}")

def train_models(self):
    print("\n--- Training Machine Learning Models ---")

    # 1. Basic Decision Tree
    dt_basic = DecisionTreeClassifier(random_state=42)
    dt_basic.fit(self.X_train, self.y_train)
    self.models['DT_Basic'] = dt_basic

    # 2. GA Optimized Decision Tree
    print("[GA] Running Genetic Algorithm Hyperparameter Search...")
    ga = GeneticAlgorithmDT(self.X_train, self.y_train, population_size=8, generations=4)
    best_params, best_fit = ga.optimize()
    print(f"[GA] Best Hyperparameters Found: {best_params} (CV F1-Score: {best_fit:.4f})")

    dt_ga = DecisionTreeClassifier(**best_params, random_state=42)
    dt_ga.fit(self.X_train, self.y_train)
    self.models['DT_GA_Optimized'] = dt_ga

```

```

# 3. Multilayer Perceptron Neural Network
print("[MLP] Training Multilayer Perceptron (Backpropagation Neural Network)...")
mlp = MLPClassifier(hidden_layer_sizes=(64, 32), max_iter=500, random_state=42,
early_stopping=True)
mlp.fit(self.X_train, self.y_train)
self.models['MLP_NeuralNet'] = mlp

def evaluate_all(self):
    print("\n===== MODEL EVALUATION SUMMARY
=====")
    results = []
    target_names = self.label_encoder.classes_

    for name, model in self.models.items():
        preds = model.predict(self.X_test)
        acc = accuracy_score(self.y_test, preds)
        f1 = f1_score(self.y_test, preds, average='weighted')
        results.append({'Model': name, 'Accuracy': acc, 'F1-Score': f1})

    print(f"\nModel: {name}")
    print(classification_report(self.y_test, preds, target_names=target_names))

    res_df = pd.DataFrame(results)
    print("Final Metric Comparison:")
    print(res_df.to_string(index=False))
    return res_df

def predict_sample(self, sample_dict):
    sample_df = pd.DataFrame([sample_dict])
    sample_proc = self.preprocessor.transform(sample_df)
    best_model = self.models['DT_GA_Optimized']

    pred_idx = best_model.predict(sample_proc)[0]
    pred_risk = self.label_encoder.inverse_transform([pred_idx])[0]

    probs = best_model.predict_proba(sample_proc)[0]
    classes = self.label_encoder.classes_

    prob_str = ", ".join([f"{c}: {p:.2%}" for c, p in zip(classes, probs)])
    return pred_risk, prob_str

```

```
# Initialize and train the system in Colab
system = CropAgriSystem()
system.load_and_preprocess()
system.train_models()
res_df = system.evaluate_all()
```

OUTPUT:

===== MODEL EVALUATION SUMMARY =====

Model: DT_Basic

	precision	recall	f1-score	support
High Risk	0.88	0.81	0.84	69
Low Risk	0.76	0.79	0.77	61
Moderate Risk	0.79	0.81	0.80	110
accuracy		0.80	240	
macro avg	0.81	0.80	0.80	240
weighted avg	0.81	0.80	0.80	240

Model: DT_GA_Optimized

	precision	recall	f1-score	support
High Risk	0.91	0.84	0.87	69
Low Risk	0.81	0.79	0.80	61
Moderate Risk	0.80	0.85	0.83	110
accuracy		0.83	240	
macro avg	0.84	0.83	0.83	240
weighted avg	0.84	0.83	0.83	240

Model: MLP_NeuralNet

	precision	recall	f1-score	support
High Risk	0.77	0.74	0.76	69
Low Risk	0.82	0.77	0.80	61

Moderate Risk	0.74	0.79	0.77	110
---------------	------	------	------	-----

accuracy		0.77	240	
macro avg	0.78	0.77	0.77	240
weighted avg	0.77	0.77	0.77	240

Final Comparison:

	Model Accuracy	F1-Score
DT_Basic	0.804167	0.804724
DT_GA_Optimized	0.833333	0.833674
MLP_NeuralNet	0.770833	0.771016